Teacher answers and topic guidance

Accept equivalent science in the usual response mode. Keep answers hidden during independent attempts.

Starter

1. Differentiation.
2. Two daughter cells with matching chromosome sets.
3. From higher to lower concentration overall.

Sort the case

Better movement is a possible benefit. Possible harmful effects are a risk. Recording harms is a further evidence check.

Argument chain

The report describes improved movement in the group. Harm results are missing. Consider further research with safeguards. Check harms and use a suitable comparison before recommending routine use.

Fresh check

Serious harms with a small movement gain weaken support. Learners can qualify or withdraw support if they connect their judgement to the evidence.

Arrival Supported

1. The first card is Movement.
2. Movement improved in the fictional study.
3. The harm result.
4. Yes.

Arrival Independent

1. Harms.
2. Improved movement in the group studied.
3. Whether participants experienced harm.
4. Record possible harmful effects.

Arrival Aspire

1. The third card, Further check.
2. Possible improvement in movement in this group.
3. Harm data were not reported.
4. Compare movement and harms across matched groups.

Arrival Choice

1. Movement.
2. Movement improved.
3. Harms.
4. Harmful effects.

Exit Supported

1. Improved movement.
2. Harm results.

Exit Independent

1. Movement improved in this group. Harms are uncertain.
2. Record possible harms or compare outcomes across groups.

Exit Aspire

1. Consider further research with safeguards because movement improved, while harms remain uncertain.
2. Serious harm or similar gains without the approach would weaken support.

Exit Choice

1. Movement improved in this group.
2. Check harms.

Made by Matt

Mark scheme Supported

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Sort these phrases as benefit, risk or evidence request: better movement, possible harm, record harms.

Better movement is the benefit. Possible harm is the risk. Record harms is the evidence request.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

2. Complete: Movement improved in the group, but the report gives no information about ___.

Harms or harmful effects.

Maximum 1 marks. Accept equivalent valid wording or a clear model demonstration.

3. Choose a next step and give a reason. Check for harms, or use the approach routinely?

Check for harms because the report does not give harm results.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt

Mark scheme Independent

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Write a claim supported by the movement result. Include who was studied.

Movement improved in the group studied. This supports a possible benefit in that group.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

2. Explain why improved movement leaves a question about safety.

Movement was reported. Harmful effects were not reported, so safety is uncertain.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

3. Recommend a next step. Link it to the evidence and the risk gap.

Consider further research because movement improved. Record harms and compare movement across groups before recommending routine use.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt

Mark scheme Aspire

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Explain how a comparison group could change the strength of the benefit claim.

Similar improvement weakens the claim that the approach caused the change. A suitable comparison helps test alternative explanations.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

2. Explain how missing harm results limit the decision, even if movement improved.

Movement data support a possible benefit. Missing harm data prevent a balanced judgement of benefits and risks.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

3. Write a qualified judgement. State what evidence would make you change it.

Consider further research with safeguards. Compare movement and harms. Serious harms or no extra movement gain would weaken support.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt

Mark scheme Choice

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Choose a supported claim: movement improved in this group, or movement improved for everyone. Give a reason.

Movement improved in this group. The report describes only that group.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

2. Choose a limit: no harms were reported, or no harms occurred. Give a reason.

No harms were reported. We cannot tell whether harms occurred.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

3. Choose a next step: check for harms, or recommend routine use. Give a reason.

Check for harms because the movement result alone leaves safety uncertain.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt